

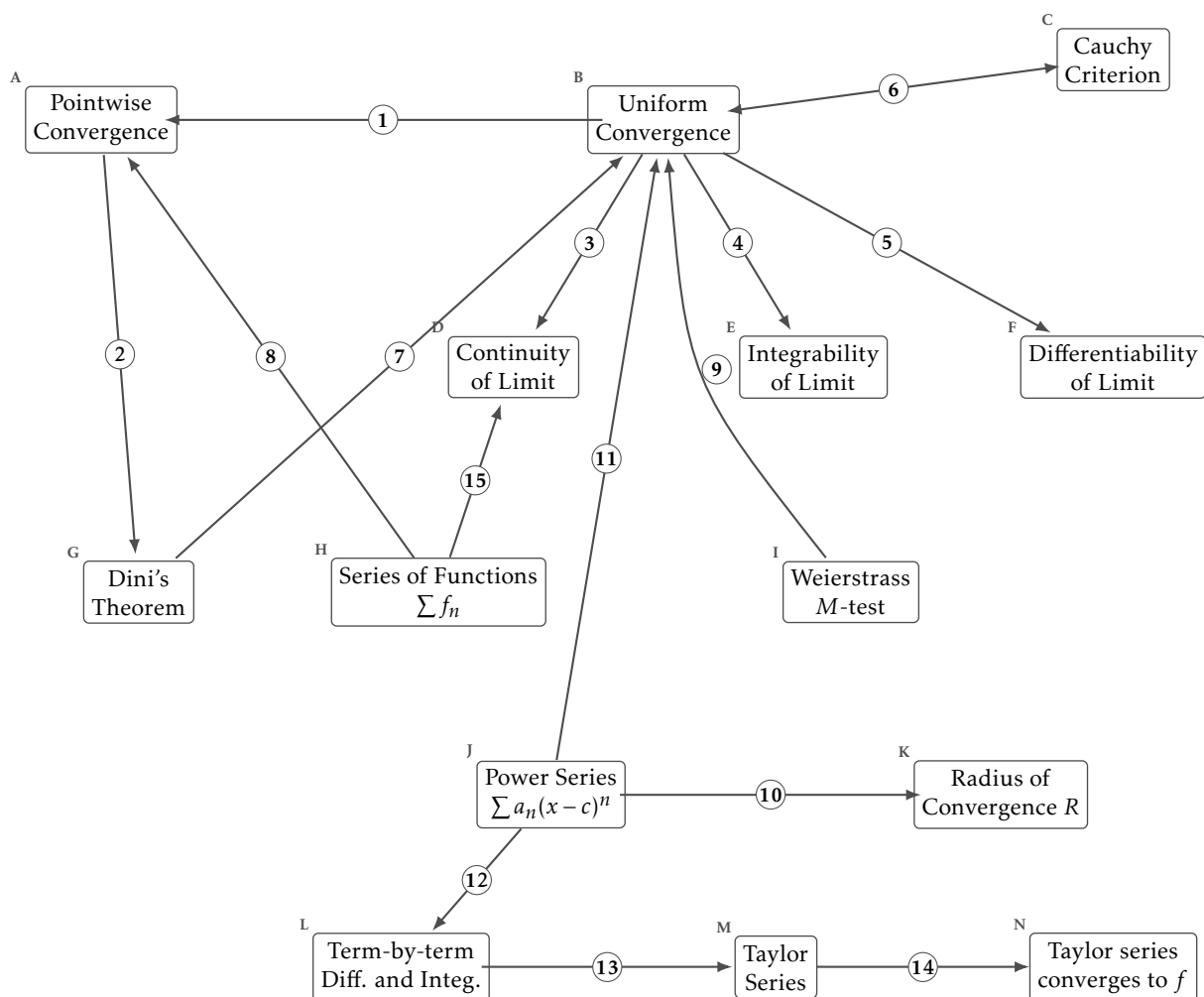
Sequences and series of functions: answers

Math 163 • Prof. Chowdhury

Every arrow, with the reason underneath. Keep this away from the blank worksheet.

What each box means

- A. Pointwise Convergence.** For each x and each $\varepsilon > 0$ there is an N depending on both, with $|f_n(x) - f(x)| < \varepsilon$ for $n \geq N$.
- B. Uniform Convergence.** For each $\varepsilon > 0$ there is an N depending only on ε , with $|f_n(x) - f(x)| < \varepsilon$ for $n \geq N$ and every x at once.
- C. Cauchy Criterion.** $\{f_n\}$ converges uniformly exactly when for each $\varepsilon > 0$ there is an N with $|f_m(x) - f_n(x)| < \varepsilon$ for all $m, n \geq N$ and all x .
- D. Continuity of Limit.** Each f_n continuous and $f_n \rightarrow f$ uniformly gives f continuous.
- E. Integrability of Limit.** Each f_n integrable and $f_n \rightarrow f$ uniformly gives f integrable, with $\lim \int f_n = \int \lim f_n$.
- F. Differentiability of Limit.** $f_n \rightarrow f$ pointwise together with $f'_n \rightarrow g$ uniformly gives $f' = g$. It is the derivatives that have to converge uniformly.
- G. Dini's Theorem.** $f_n \rightarrow f$ pointwise on $[a, b]$, monotonely, with every f_n and f continuous, gives $f_n \rightarrow f$ uniformly.
- H. Series of Functions** $\sum f_n$. $\sum f_n$ converges when its partial sums $S_N(x) = \sum_{k \leq N} f_k(x)$ do. A series of functions is a sequence of functions.
- I. Weierstrass M-test.** If $|f_n(x)| \leq M_n$ for every x and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly and absolutely.
- J. Power Series** $\sum a_n(x - c)^n$. A series of functions whose n th term is a monomial of degree n centred at c .
- K. Radius of Convergence** R . $\sum a_n(x - c)^n$ converges absolutely for $|x - c| < R$ and diverges for $|x - c| > R$. The endpoints are checked one at a time.
- L. Term-by-term Diff. and Integ..** Inside $(c - R, c + R)$ a power series can be differentiated and integrated term by term, and the result has the same radius R .
- M. Taylor Series.** $\sum f^{(n)}(a)(x - a)^n/n!$. The coefficients are forced by differentiating the power series at its centre.
- N. Taylor series converges to f .** The Taylor series converges to $f(x)$ exactly when the remainder $R_N(x) = f(x) - P_N(x)$ tends to zero.



The arrows

1. Uniform convergence is pointwise convergence with an N that does not depend on x , so it is the stronger of the two. One quantifier moves and everything in the chapter follows from where it lands. Pointwise lets N chase x ; uniform does not.
2. Dini's theorem starts from pointwise convergence and adds compactness, monotone convergence and a continuous limit, and gets uniform convergence out. Pointwise convergence on its own carries nothing to the limit: every x^n on $[0, 1]$ is continuous and the limit is not. Dini's three extra hypotheses are what close that gap, and dropping any one of them reopens it.
3. A uniform limit of continuous functions is continuous. The three-epsilon argument: get near f_n uniformly, use the continuity of that one f_n , and come back.
4. A uniform limit of integrable functions is integrable, and the limit of the integrals is the integral of the limit. Uniform closeness bounds the difference of the integrals by the length of the interval times ϵ , which is what lets the two operations swap.
5. The differentiability theorem is uniform convergence imposed on the derivatives rather than on the functions: $f_n \rightarrow f$ pointwise with $f'_n \rightarrow g$ uniformly gives $f' = g$. It has to be the derivatives. $\sqrt{x^2 + 1/n^2} \rightarrow |x|$ uniformly and $|x|$ is not differentiable at 0, and $(1/n)\sin(n^2x) \rightarrow 0$ uniformly while its derivatives diverge. After the clean results for continuity and integrability, this one is disappointing.
6. A sequence of functions converges uniformly exactly when it is uniformly Cauchy, so neither statement needs the limit named in advance. This is what makes the M -test possible: you can prove uniform convergence of a series without ever writing down its sum.
7. On a closed bounded interval, monotone pointwise convergence of continuous functions to a continuous limit is already uniform. Every hypothesis is doing work. Drop compactness, monotonicity, or the continuity of the limit and there is a counterexample waiting.
8. A series of functions is the sequence of its partial sums, so every statement about sequences of functions applies to it unchanged. Nothing about series of functions is new. It is the sequence $\{S_N\}$ wearing different notation.
9. Bounding each term by a constant whose series converges gives uniform and absolute convergence of the series of functions. The bound has to hold for every x with one constant per term, which is exactly how the x is got out of the way.
10. A power series converges absolutely inside one radius and diverges outside it, and that radius is what the ratio or root test computes. The trichotomy is the theorem. What happens at the two endpoints is decided separately and can go either way.
11. Inside its radius a power series converges uniformly on every closed bounded subinterval, which is what licenses working on it term by term. Uniform on compact subsets, not on the whole open interval. That distinction is why the statement names a closed bounded subinterval.
12. Because the convergence is uniform on compact subintervals, a power series can be differentiated and integrated term by term, and the result keeps the same radius. The interchange theorems supply this, and the radius is unchanged because the ratio or root test sees the same limit for the derived series.
13. Differentiating term by term at the centre forces the coefficients to be $f^{(n)}(c)/n!$, so a function has at most one power series about a point. Uniqueness comes free. Any two power series agreeing near a point have the same coefficients, because both sets are computed from the same derivatives.
14. The Taylor series converges to f at a point exactly when the remainder tends to zero there, which is a statement about f and not about the coefficients. A series can have every Taylor coefficient of f , converge everywhere, and still not converge to f . The remainder is what decides it.
15. Applying the interchange theorems to the partial sums gives the matching statements for the sum of a series of functions. A uniformly convergent series of continuous functions has a continuous sum, and can be integrated term by term, for exactly the reason the sequence version holds.

Claims that look right and are not

- A uniform limit of differentiable functions is differentiable.
- A power series and the series obtained by differentiating it term by term have different radii of convergence.
- A function with derivatives of every order at a point is the sum of its Taylor series near that point.
- Pointwise convergence on a closed bounded interval is automatically uniform.
- A series of continuous functions that converges pointwise has a continuous sum.

Counterexamples and benchmarks

- x^n on $[0, 1]$ takes continuous functions to a discontinuous limit, pointwise.
- $\sqrt{x^2 + 1/n^2} \rightarrow |x|$ uniformly, and the limit is not differentiable at 0.
- $(1/n) \sin(n^2 x) \rightarrow 0$ uniformly while its derivatives diverge.
- $x/(1 + n^2 x^2) \rightarrow 0$ uniformly, and the derivative of the limit is not the limit of the derivatives at 0.
- e^{-1/x^2} extended by 0 is infinitely differentiable at 0 with every Taylor coefficient zero, so its Taylor series converges and not to it.

One last question

Which single arrow carries the most of this chapter? One quantifier separates the two kinds of convergence, and most of the chapter follows from where it lands, so say whether your answer is the arrow that moves it.